

Air Conditioners Data Analysis

Stage 7: Data Visualization

This notebook is independently executable in Google Colab. Every step explains what we do, why we do it, what we obtain, and how the result is used in the next stage.

Stage Objective

Stage 7 communicates the analysis visually so patterns can be understood faster than from raw tables alone.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load the project CSV directly so this notebook is independently executable.
csv_path = "/content/Air_Conditioners_Cleaned.csv"
df = pd.read_csv(csv_path)

print("CSV loaded successfully.")
print("Dataset shape:", df.shape)
```

```
CSV loaded successfully.
Dataset shape: (720, 14)
```

Why Visualisation

Charts make distributions, category differences and relationships easier to compare and help the audience see the evidence behind the analysis.

```
nums=df.select_dtypes(include=np.number).columns.tolist(); cats=df.select_dtypes(include=['object','category','bool']).columns.tolist(); print(nums,cats)

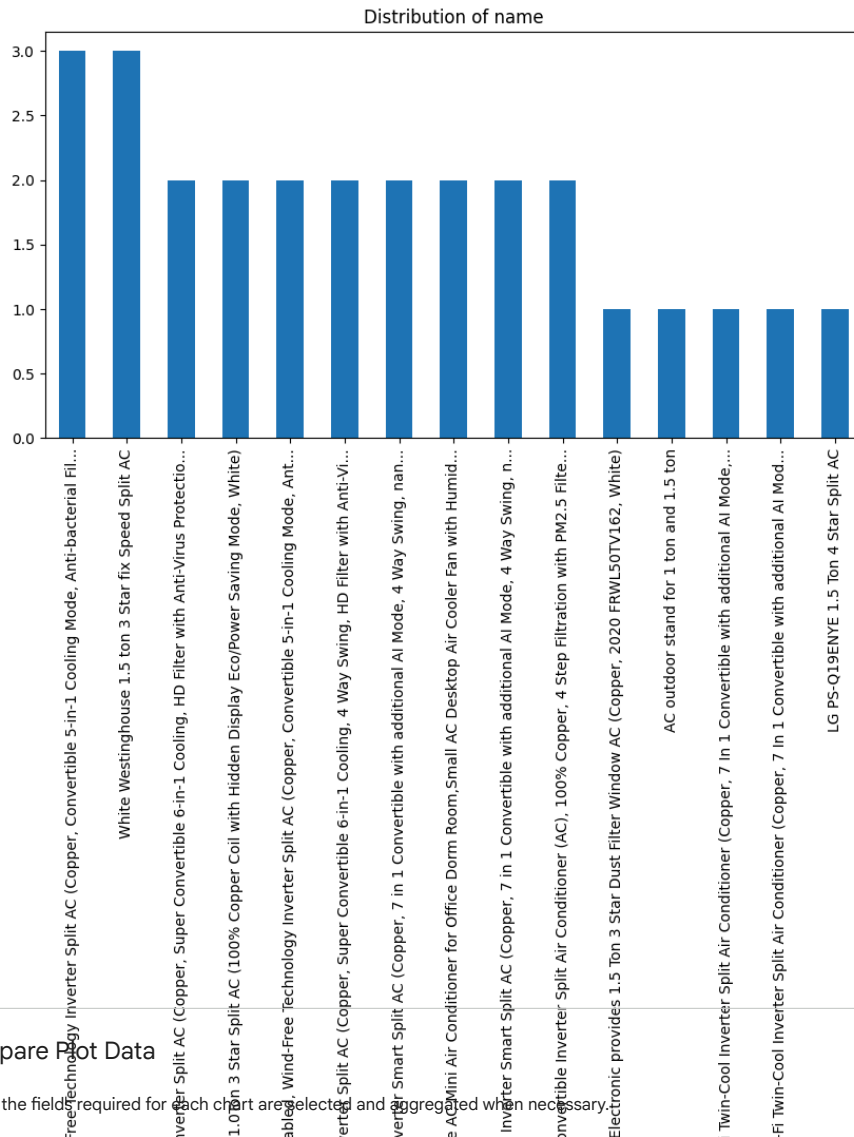
['ratings', 'no_of_ratings', 'discount_price', 'actual_price', 'discount_amount', 'discount_percent', 'tonnage', 'energy_star'] ['name', 'main_category', 'sub_ca
```

Reload the CSV

The CSV is loaded directly so the visualisation stage can be run independently.

```
if cats:
    df[cats[0]].value_counts().head(15).plot(kind='bar',figsize=(10,5),title=f'Distribution of {cats[0]}'); plt.tight_layout(); plt.show()
```

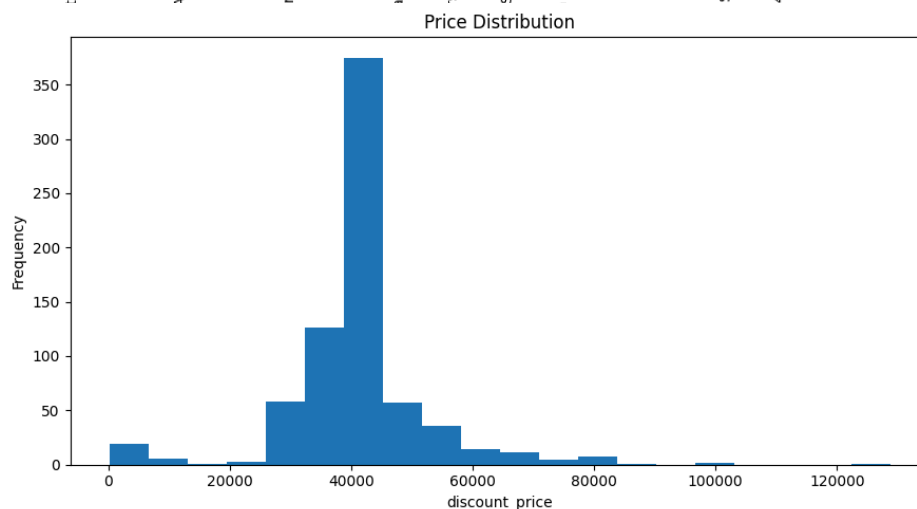
```
/tmp/ipykernel_487/3160831646.py:2: UserWarning: Tight layout not applied. The bottom and top margins cannot be made large enough to accommodate all Axes decor
df[cats[0]].value_counts().head(15).plot(kind='bar',figsize=(10,5),title=f'Distribution of {cats[0]}'); plt.tight_layout(); plt.show()
```



Prepare Plot Data

Only the fields required for each chart are selected and aggregated when necessary.

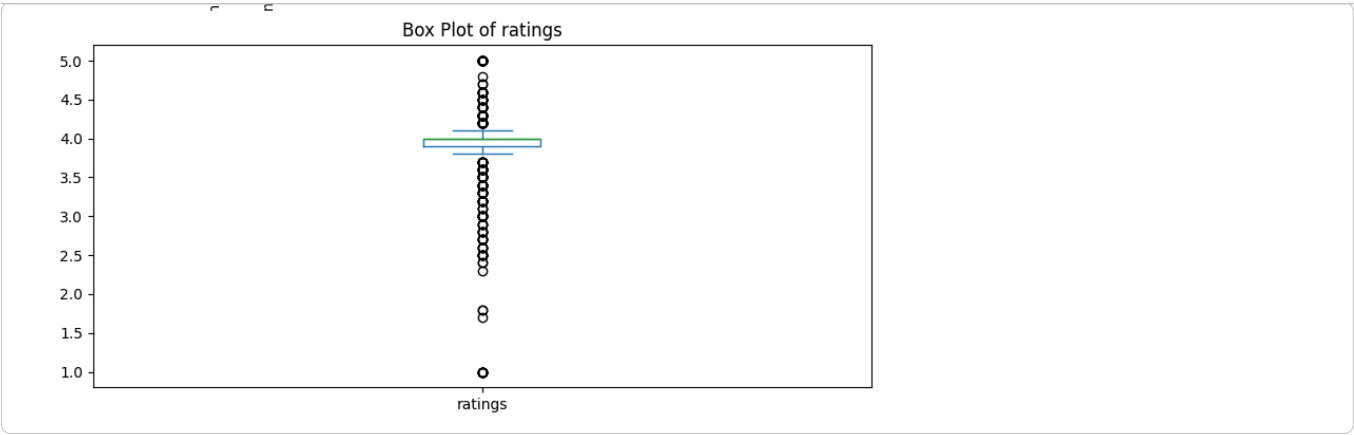
```
if 'discount_price' in df.columns:
df['discount_price'].plot(kind='hist',bins=20,figsize=(9,5),title='Price Distribution'); plt.xlabel('discount_price'); plt.tight_layout(); plt.show()
```



Category Distribution

A bar chart is used for a categorical distribution because bar charts make differences in category counts easy to compare.

```
if 'ratings' in df.columns:
df['ratings'].plot(kind='hist',bins=10,figsize=(9,5),title='Rating Distribution'); plt.xlabel('ratings'); plt.tight_layout(); plt.show()
```

Group Comparison

A grouped summary is visualised to compare a numerical measure across product categories.

